

Comunicações Móveis

5G Open5GS

Eduardo Alves, nº104179

Bernardo Marujo, nº107322

Daniel Calónico, nº134953

André Alexandre, nº114143

Mestrado em Engenharia de Computadores e Telemática

Universidade de Aveiro



Objetivos

Implementação de uma arquitetura 5G Standalone baseada em Open5GS.

- Deployment do core.

Duas soluções diferentes:

Namespaces

Docker

- Emulação da RAN usando UERANSIM.
- Network Slicing
- QoS Policies
- Análise do Tráfego com várias topologias de rede.

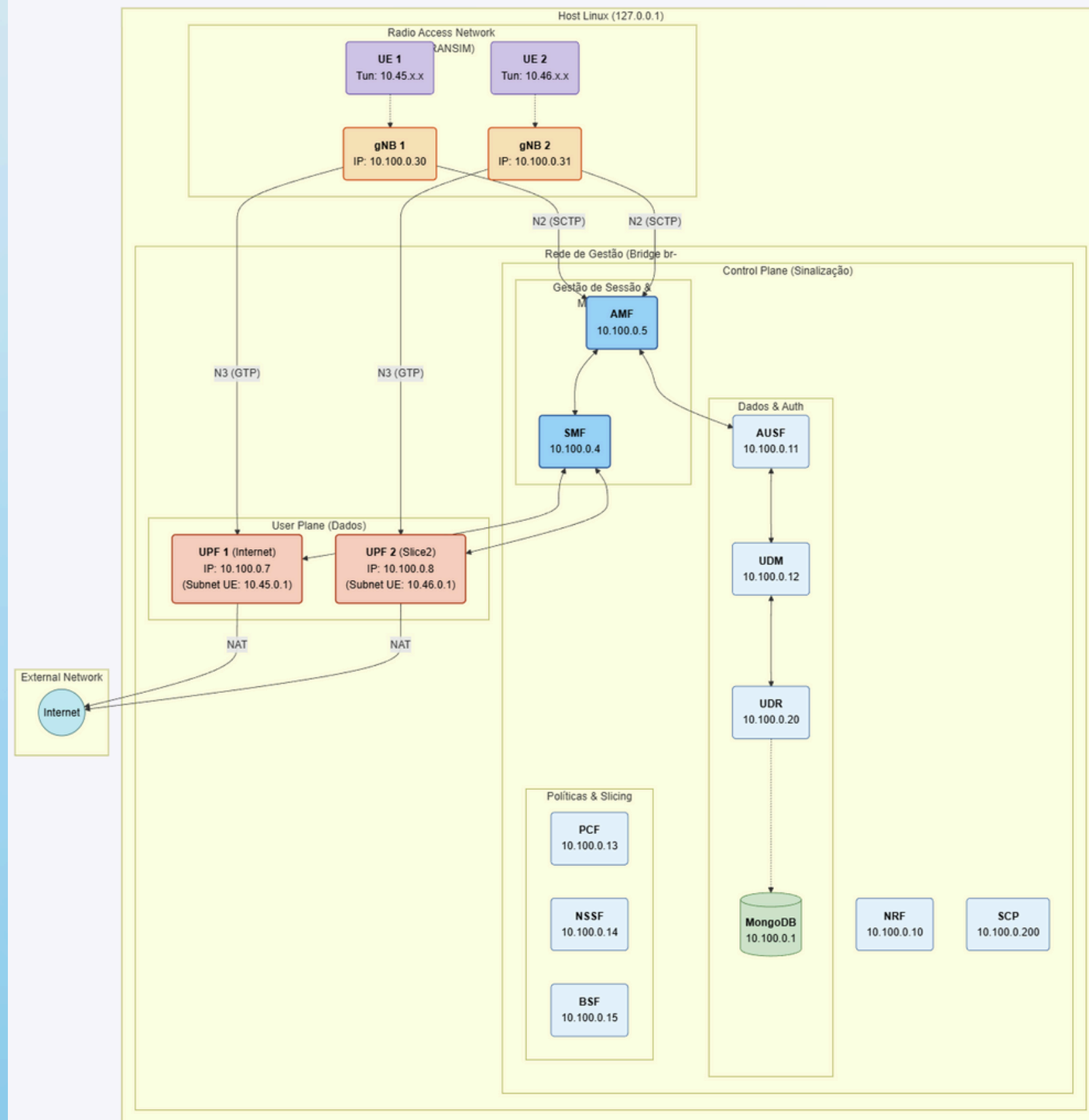


Ferramentas usadas

- Open5GS (Core)
 MongoDB
 WebUI
- UERANSIM (RAN)
- Docker
- Namespaces
- Bash (Scripts para automatização)



Arquitetura (Namespaces)

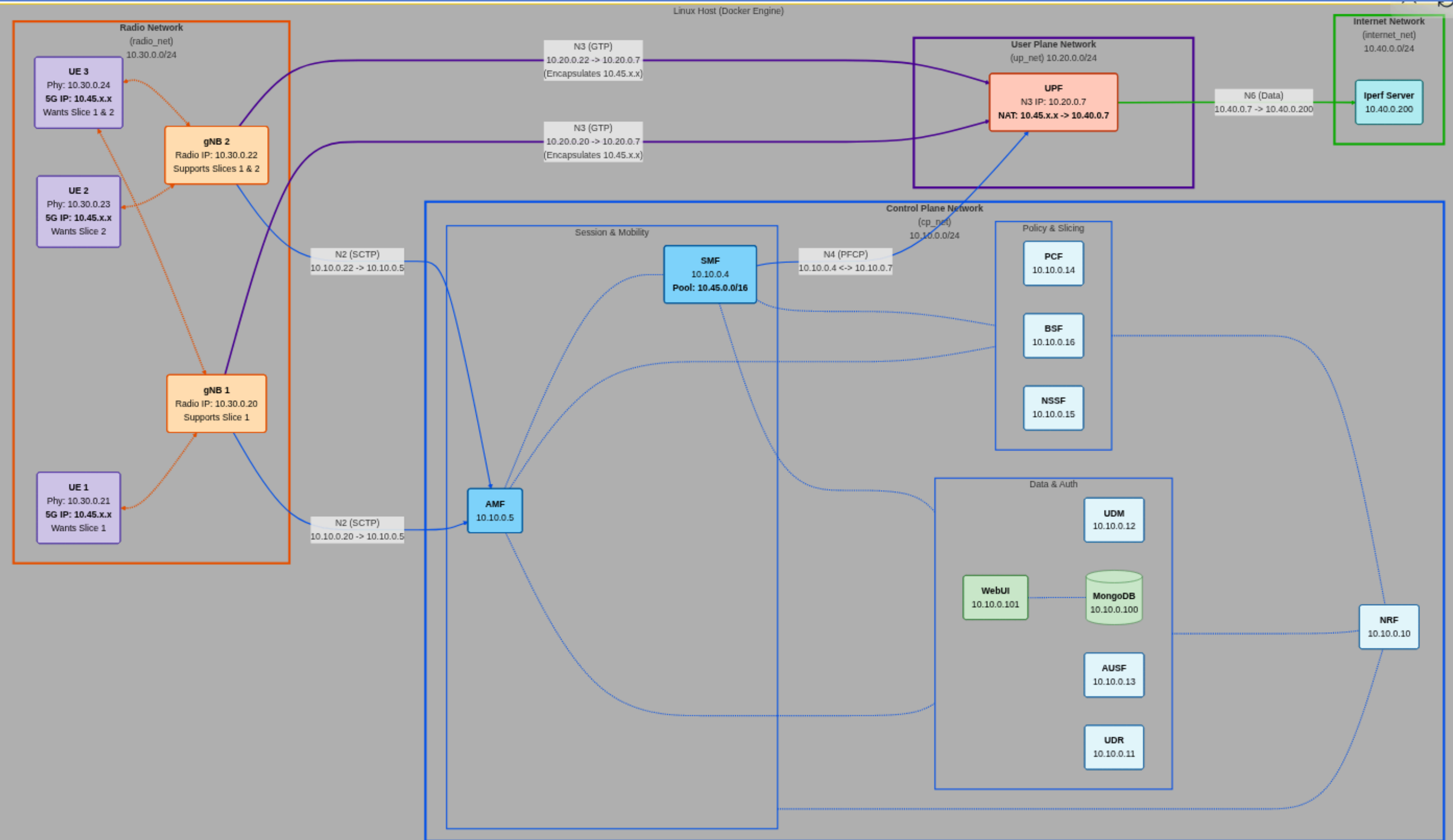


Configurações (Namespaces)

```
# =====  
# 3. Criar Namespaces para o Core 5G  
# =====  
  
create_ns ns-nrf 10.100.0.10  
create_ns ns-amf 10.100.0.5  
create_ns ns-smf 10.100.0.4  
create_ns ns-upf1 10.100.0.7  
create_ns ns-upf2 10.100.0.8  
create_ns ns-udr 10.100.0.20  
create_ns ns-udm 10.100.0.12  
create_ns ns-pcf 10.100.0.13  
create_ns ns-ausf 10.100.0.11  
create_ns ns-nssf 10.100.0.14  
create_ns ns-bsf 10.100.0.15  
create_ns ns-scp 10.100.0.200  
  
# =====  
# 4. Criar Namespaces para RAN  
# =====  
  
create_ns ns-gnb1 10.100.0.30  
create_ns ns-gnb2 10.100.0.31
```

```
create_ns() {  
    NS=$1  
    IP=$2  
    echo "A configurar namespace $NS com IP $IP..."  
  
    # 1. Garantir que o namespace está limpo antes de começar  
    ip netns del $NS 2>/dev/null  
    ip netns add $NS  
  
    # 2. Criar par veth com nomes ÚNICOS para evitar conflitos  
    # v-$NS fica no host, p-$NS é o peer temporário  
    ip link add v-$NS type veth peer name p-$NS  
  
    # 3. Ligar interface do host à bridge  
    ip link set v-$NS master br-5g  
    ip link set v-$NS up  
  
    # 4. Mover o peer ÚNICO para o namespace  
    ip link set p-$NS netns $NS  
  
    # 5. Renomear para eth0 DENTRO do namespace e configurar IP  
    ip netns exec $NS ip link set dev p-$NS name eth0  
    ip netns exec $NS ip addr add $IP/24 dev eth0  
    ip netns exec $NS ip link set eth0 up  
    ip netns exec $NS ip link set lo up  
  
    # 6. Definir Gateway  
    ip netns exec $NS ip route add default via 10.100.0.1  
}
```

Arquitetura (Docker)



Configurações (Docker)

```
networks:
  cp_net:
    driver: bridge
    ipam:
      config:
        - subnet: 10.10.0.0/24
    driver_opts:
      com.docker.network.bridge.name: br-cp
  up_net:
    driver: bridge
    ipam:
      config:
        - subnet: 10.20.0.0/24
    driver_opts:
      com.docker.network.bridge.name: br-up
  radio_net:
    driver: bridge
    ipam:
      config:
        - subnet: 10.30.0.0/24
    driver_opts:
      com.docker.network.bridge.name: br-radio
  internet_net:
    driver: bridge
    ipam:
      config:
        - subnet: 10.40.0.0/24
    driver_opts:
      com.docker.network.bridge.name: br-internet
```

```
upf:
  image: gradiant/open5gs:2.7.5
  container_name: upf
  hostname: upf
  user: root
  privileged: true
  restart: on-failure
  cap_add:
    - all
  volumes:
    - ./configs/upf.yaml:/opt/open5gs/etc/open5gs/upf.yaml
  sysctls:
    - net.ipv4.ip_forward=1
  entrypoint: /bin/sh
  command:
    - -c
    - |
      ip tuntap add name ogstun mode tun
      ip addr add 10.45.0.1/16 dev ogstun
      ip addr add 2001:db8:cafe::1/48 dev ogstun
      ip link set ogstun up
      iptables -t nat -A POSTROUTING -s 10.45.0.0/16 ! -o ogstun -j MASQUERADE
      exec open5gs-upfd -c /opt/open5gs/etc/open5gs/upf.yaml
  dns:
    - 8.8.8.8
    - 8.8.4.4
  networks:
    cp_net:
      ipv4_address: 10.10.0.7
    up_net:
      ipv4_address: 10.20.0.7
    internet_net:
      ipv4_address: 10.40.0.7
```

Configurações (Docker)

```
ue1:
  image: gradient/ueransim:3.2.7
  container_name: ue1
  hostname: ue1
  privileged: true
  restart: on-failure
  cap_add:
    - all
  depends_on: [gnb1]
  volumes:
    - ./configs/ue1.yaml:/ueransim/config/ue.yaml
  entrypoint: /bin/sh
  command:
    - -c
    - nr-ue -c /ueransim/config/ue.yaml
  dns:
    - 8.8.8.8
    - 8.8.4.4
  networks:
    radio_net:
      ipv4_address: 10.30.0.21
```

```
amf:
  image: gradient/open5gs:2.7.5
  container_name: amf
  hostname: amf
  command: open5gs-amfd
  restart: on-failure
  depends_on: [nrf]
  volumes:
    - ./configs/amf.yaml:/opt/open5gs/etc/open5gs/amf.yaml
  dns:
    - 8.8.8.8
    - 8.8.4.4
  networks:
    cp_net:
      ipv4_address: 10.10.0.5
```

```
gnb1:
  image: gradient/ueransim:3.2.7
  container_name: gnb1
  hostname: gnb1
  privileged: true
  restart: on-failure
  depends_on: [amf]
  volumes:
    - ./configs/gnb1.yaml:/ueransim/config/gnb.yaml
  entrypoint: /bin/sh
  command:
    - -c
    - nr-gnb -c /ueransim/config/gnb.yaml
  dns:
    - 8.8.8.8
    - 8.8.4.4
  networks:
    cp_net:
      ipv4_address: 10.10.0.20
    up_net:
      ipv4_address: 10.20.0.20
    radio_net:
      ipv4_address: 10.30.0.20
```

Network Slicing

Subscriber 1

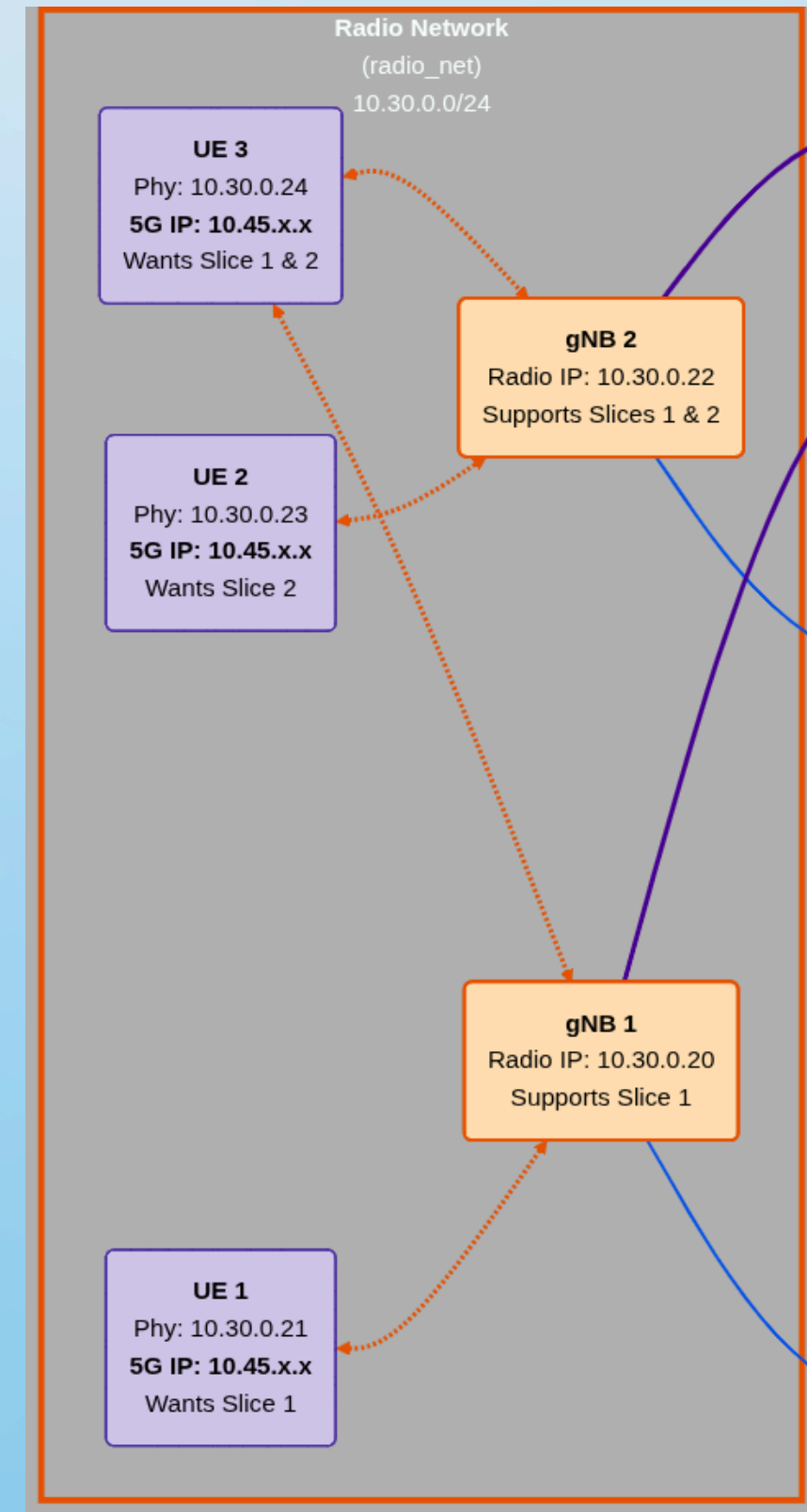
- **IMSI:** 9997000000000001
- **K:** 465B5CE8B199B49FAA5F0A2EE238A6BC
- **OPc:** E8ED289DEBA952E4283B54E88E6183CA
- **opType:** OPc
- **Slice:** SST 1, SD 000001

Subscriber 2

- **IMSI:** 9997000000000002
- **K:** 465B5CE8B199B49FAA5F0A2EE238A6BD
- **OPc:** E8ED289DEBA952E4283B54E88E6183CB
- **opType:** OPc
- **Slice:** SST 2, SD 000002

Subscriber 3

- **IMSI:** 9997000000000003
- **K:** 465B5CE8B199B49FAA5F0A2EE238A6BE
- **OPc:** E8ED289DEBA952E4283B54E88E6183CC
- **opType:** OPc
- **Slice:** SST 1, SD 000001 and SST 2, SD 000002



Network Slicing

9997000000000003

4370816125816151 ...IMEISV

465B5CE8B199B49FAA5F0A2EE238A6BE ...K

E8ED289DEBA952E4283B54E88E6183CC ...OPc

8000 ...AMF

737 ...SQN

SERVICE_GRANTED (0) ...
Subscriber Status (TS 29.272 7.3.29)

0 ...Operator Determined Barring
(TS 29.272 7.3.30)

1 Gbps ...DL

1 Gbps ...UL

SST:1 SD:000001 (Default S-NSSAI)

DNN/APN	Type	5QI/QCI	ARP	Capability	Vulnerability	MBR DL/UL	GBR DL/UL
internet	IPv4	9	8	Disabled	Disabled	2 Mbps / 2 Mbps	

SST:2 SD:000002 (Default S-NSSAI)

DNN/APN	Type	5QI/QCI	ARP	Capability	Vulnerability	MBR DL/UL	GBR DL/UL
internet	IPv4	5	1	Disabled	Disabled	1 Gbps / 1 Gbps	

Network Slicing

ngap					
No.	Time	Source	Destination	Protocol	Length Info
48	30.440868948	10.10.0.22	10.10.0.5	NGAP/N...	138 InitialUEMessage, Registration request [RRCEstablishmentCause=mo-Signalling]
94	30.451946172	10.10.0.5	10.10.0.22	NGAP/N...	146 SACK (Ack=0, Arwnd=106496), DownlinkNASTransport, Authentication request
95	30.452941409	10.10.0.22	10.10.0.5	NGAP/N...	146 SACK (Ack=0, Arwnd=106496), UplinkNASTransport, Authentication response
102	30.457608602	10.10.0.5	10.10.0.22	NGAP/N...	126 SACK (Ack=1, Arwnd=106496), DownlinkNASTransport, Security mode command
103	30.458451265	10.10.0.22	10.10.0.5	NGAP/N...	194 SACK (Ack=1, Arwnd=106496), UplinkNASTransport
104	30.458862191	10.10.0.5	10.10.0.22	NGAP	98 SACK (Ack=2, Arwnd=106496), UEContextReleaseCommand [Cause: Nas=normal-release]
106	30.459047696	10.10.0.22	10.10.0.5	NGAP	122 SACK (Ack=2, Arwnd=106496), UEContextReleaseComplete
162	30.470316308	10.10.0.5	10.10.0.22	NGAP/N...	246 SACK (Ack=3, Arwnd=106496), InitialContextSetupRequest
163	30.470608552	10.10.0.22	10.10.0.5	NGAP	98 SACK (Ack=3, Arwnd=106496), InitialContextSetupResponse
167	30.672099051	10.10.0.22	10.10.0.5	NGAP/N...	362 UplinkNASTransport, UplinkNASTransport, UplinkNASTransport
168	30.672685226	10.10.0.5	10.10.0.22	NGAP/N...	154 SACK (Ack=7, Arwnd=106395), DownlinkNASTransport
198	30.688493793	10.10.0.4	10.10.0.5	HTTP2/...	755 DATA[1], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
203	30.688960473	10.10.0.4	10.10.0.5	HTTP2/...	755 DATA[3], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
206	30.689125833	10.10.0.5	10.10.0.22	NGAP/N...	250 PDUSessionResourceSetupRequest
209	30.689447737	10.10.0.5	10.10.0.22	NGAP/N...	234 PDUSessionResourceSetupRequest
213	30.691831055	10.10.0.22	10.10.0.5	NGAP	142 SACK (Ack=6, Arwnd=106496), PDUSessionResourceSetupResponse
215	30.692226059	10.10.0.5	10.10.0.4	HTTP2/...	353 DATA[9], JSON (application/json), PDUSessionResourceSetupResponseTransfer
220	30.896029552	10.10.0.22	10.10.0.5	NGAP	126 PDUSessionResourceSetupResponse
222	30.896876304	10.10.0.5	10.10.0.4	HTTP2/...	353 DATA[11], JSON (application/json), PDUSessionResourceSetupResponseTransfer

value

PDUSessionResourceSetupRequest

protocolIEs: 4 items

Item 0: id-AMF-UE-NGAP-ID

ProtocolIE-Field

id: id-AMF-UE-NGAP-ID (10)

criticality: reject (0)

value

AMF-UE-NGAP-ID: 5

Item 1: id-RAN-UE-NGAP-ID

ProtocolIE-Field

id: id-RAN-UE-NGAP-ID (85)

criticality: reject (0)

value

RAN-UE-NGAP-ID: 3

Item 2: id-PDUSessionResourceSetupListSUReq

ProtocolIE-Field

id: id-PDUSessionResourceSetupListSUReq (74)

criticality: reject (0)

value

PDUSessionResourceSetupListSUReq: 1 item

Item 0

PDUSessionResourceSetupItemSUReq

pduSessionID: 1

pduSessionNAS-PDU: 7e02ce8b5a07037e006801004a2e0101c2110009010006:

s-NSSAI

sST: 01

sD: 000001

pduSessionResourceSetupRequestTransfer: 0000040082000a0c3b9aca003:

Item 3: id-UEAggregateMaximumBitRate

0000 86 46 ec fc 5b ef 22 35 87 c4 65 10 08 00 45 02 .F..["5..e...E.

0010 00 ec 1d 4e 40 00 40 84 08 10 0a 0a 00 05 0a 0a ...N@.....

0020 00 16 96 0c db 19 c5 eb cb 0f 00 00 00 00 00 03
0030 00 c9 6a 1b 1c aa 00 03 00 04 00 00 00 3c 00 1d ..j.....<..
0040 00 80 b4 00 00 04 00 0a 00 02 00 05 00 55 00 02U..
0050 00 03 00 4a 00 80 92 00 40 01 59 7e 02 ce 8b 5a ...J....@.Y...Z
0060 07 03 7e 00 68 01 00 4a 2e 01 01 c2 11 00 09 01 ...~h..J.....
0070 00 06 31 31 01 01 ff 01 06 03 f4 24 03 f4 24 29 ..11....\$..\$)
0080 05 01 0a 2d 00 07 22 04 01 00 00 01 79 00 06 01". ...y..
0090 20 41 01 01 09 7b 00 0f 80 00 0d 04 08 08 08 08 A...{.....
00a0 00 0d 04 08 08 04 04 25 09 08 69 6e 74 65 72 6e% ..intern
00b0 65 74 12 01 40 20 00 00 01 2f 00 00 04 00 82 00 et..@ .. /.....
00c0 0a 0c 3b 9a ca 00 30 3b 9a ca 00 00 8b 00 0a 01 ..;...0;
00d0 f0 0a 14 00 07 00 00 6a d9 00 86 00 01 00 00 88j
00e0 00 07 00 01 00 00 09 1c 00 00 6e 40 0a 0c 3b 9a-n@...;
00f0 ca 00 30 3b 9a ca 00 00 00 000;....

Packet (250 bytes) Unaligned OCTET STRING (1 byte)

Network Slicing

Time	Source	Destination	Protocol	Length	Info
48	30.440868948	10.10.0.22	10.10.0.5	NGAP/N...	138 InitialUEMessage, Registration request [RRCEstablishmentCause=mo-Signalling]
94	30.451946172	10.10.0.5	10.10.0.22	NGAP/N...	146 SACK (Ack=0, Arwnd=106496) , DownlinkNASTransport, Authentication request
95	30.452941409	10.10.0.22	10.10.0.5	NGAP/N...	146 SACK (Ack=0, Arwnd=106496) , UplinkNASTransport, Authentication response
102	30.457608602	10.10.0.5	10.10.0.22	NGAP/N...	126 SACK (Ack=1, Arwnd=106496) , DownlinkNASTransport, Security mode command
103	30.458451265	10.10.0.22	10.10.0.5	NGAP/N...	194 SACK (Ack=1, Arwnd=106496) , UplinkNASTransport
104	30.458862191	10.10.0.5	10.10.0.22	NGAP	98 SACK (Ack=2, Arwnd=106496) , UEContextReleaseCommand [Cause: Nas=normal-release]
106	30.459047696	10.10.0.22	10.10.0.5	NGAP	122 SACK (Ack=2, Arwnd=106496) , UEContextReleaseComplete
162	30.470316308	10.10.0.5	10.10.0.22	NGAP/N...	246 SACK (Ack=3, Arwnd=106496) , InitialContextSetupRequest
163	30.470608552	10.10.0.22	10.10.0.5	NGAP	98 SACK (Ack=3, Arwnd=106496) , InitialContextSetupResponse
167	30.672099051	10.10.0.22	10.10.0.5	NGAP/N...	362 UplinkNASTransport, UplinkNASTransport, UplinkNASTransport
168	30.672685226	10.10.0.5	10.10.0.22	NGAP/N...	154 SACK (Ack=7, Arwnd=106395) , DownlinkNASTransport
198	30.688493793	10.10.0.4	10.10.0.5	HTTP2/_	755 DATA[1], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
203	30.688960473	10.10.0.4	10.10.0.5	HTTP2/_	755 DATA[3], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
206	30.689125833	10.10.0.5	10.10.0.22	NGAP/N...	250 PDUSessionResourceSetupRequest
209	30.689447737	10.10.0.5	10.10.0.22	NGAP/N...	234 PDUSessionResourceSetupRequest
213	30.691831055	10.10.0.22	10.10.0.5	NGAP	142 SACK (Ack=6, Arwnd=106496) , PDUSessionResourceSetupResponse
215	30.692226059	10.10.0.5	10.10.0.4	HTTP2/_	353 DATA[9], JSON (application/json), PDUSessionResourceSetupResponseTransfer
220	30.896029552	10.10.0.22	10.10.0.5	NGAP	126 PDUSessionResourceSetupResponse
222	30.896876304	10.10.0.5	10.10.0.4	HTTP2/_	353 DATA[11], JSON (application/json), PDUSessionResourceSetupResponseTransfer

criticality: reject (0)

value

PDUSessionResourceSetupRequest

protocolIEs: 3 items

Item 0: id-AMF-UE-NGAP-ID

ProtocolIE-Field

id: id-AMF-UE-NGAP-ID (10)

criticality: reject (0)

value

AMF-UE-NGAP-ID: 5

Item 1: id-RAN-UE-NGAP-ID

ProtocolIE-Field

id: id-RAN-UE-NGAP-ID (85)

criticality: reject (0)

value

RAN-UE-NGAP-ID: 3

Item 2: id-PDUSessionResourceSetupListSUReq

ProtocolIE-Field

id: id-PDUSessionResourceSetupListSUReq (74)

criticality: reject (0)

value

PDUSessionResourceSetupListSUReq: 1 item

Item 0

PDUSessionResourceSetupItemSUReq

pduSessionID: 2

pduSessionNAS-PDU: 7e028da521dd047e006801004a2e0202c2110009010006

s-NSSAI

sST: 02

sD: 000002

pduSessionResourceSetupRequestTransfer: 0000040082000a0c3b9aca003

0000

86 46 ec fc 5b ef 22 35 87 c4 65 10 08 00 45 02

0010

00 dc 1d 4f 40 00 40 84 08 1f 0a 0a 00 05 0a 0a

0020

00 16 96 0c db 19 c5 eb cb 0f 00 00 00 00 00 03

0030

00 bb 6a 1b 1c ab 00 03 00 05 00 00 00 3c 00 1d

0040

00 80 a6 00 00 03 00 0a 00 02 00 05 00 55 00 02

0050

00 03 00 4a 00 80 92 00 40 02 59 7e 02 8d a5 21

0060

dd 04 7e 00 68 01 00 4a 2e 02 02 c2 11 00 09 01

0070

00 06 31 31 01 01 ff 01 06 03 f4 24 03 f4 24 29

0080

05 01 0a 2d 00 08 22 04 02 00 00 02 79 00 06 01

0090

20 41 01 01 05 7b 00 0f 80 00 0d 04 08 08 08 08

00a0

00 0d 04 08 08 04 04 25 09 08 69 6e 74 65 72 6e

00b0

65 74 12 02 40 40 00 00 02 2f 00 00 04 00 82 00

00c0

0a 0c 3b 9a ca 00 30 3b 9a ca 00 00 8b 00 0a 01

00d0

f0 0a 14 00 07 00 00 88 92 00 86 00 01 00 00 88

00e0

00 07 00 01 00 00 05 00 00 00

Packet (324 bytes) Unaligned OCTET STRING (1 byte)

Network Slicing

142	16.169155387	10.10.0.22	10.10.0.5	NGAP/N...	362 UplinkNASTransport, UplinkNASTransport, UplinkNASTransport
143	16.169991166	10.10.0.5	10.10.0.22	NGAP/N...	154 SACK (Ack=7, Arwnd=106395) , DownlinkNASTransport
170	16.184527658	10.10.0.4	10.10.0.5	HTTP2/...	755 DATA[1], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
179	16.184992327	10.10.0.4	10.10.0.5	HTTP2/...	753 DATA[3], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
180	16.184992623	10.10.0.5	10.10.0.22	NGAP/N...	250 PDUSessionResourceSetupRequest
185	16.185375024	10.10.0.5	10.10.0.22	NGAP/N...	234 PDUSessionResourceSetupRequest
188	16.187743620	10.10.0.22	10.10.0.5	NGAP	142 SACK (Ack=6, Arwnd=106496) , PDUSessionResourceSetupResponse
190	16.188099758	10.10.0.5	10.10.0.4	HTTP2/...	353 DATA[9], JSON (application/json), PDUSessionResourceSetupResponseTransfer
195	16.392936756	10.10.0.22	10.10.0.5	NGAP	126 PDUSessionResourceSetupResponse
197	16.393350412	10.10.0.5	10.10.0.4	HTTP2/...	353 DATA[11], JSON (application/json), PDUSessionResourceSetupResponseTransfer

```
  ▾ PDUSessionAggregateMaximumBitRate
    pDUSessionAggregateMaximumBitRateDL: 20000000bits/s
    pDUSessionAggregateMaximumBitRateUL: 20000000bits/s
  ▸ Item 1: id-UL-NGU-UP-TNLIInformation
  ▸ Item 2: id-PDUSessionType
  ▾ Item 3: id-QosFlowSetupRequestList
    ▾ ProtocolIE-Field
      id: id-QosFlowSetupRequestList (136)
      criticality: reject (0)
      ▾ value
        ▾ QosFlowSetupRequestList: 1 item
          ▾ Item 0
            ▾ QosFlowSetupRequestItem
              qosFlowIdentifier: 1
              ▾ qosFlowLevelQosParameters
                ▾ qosCharacteristics: nonDynamic5QI (0)
                  ▾ nonDynamic5QI
                    fiveQI: 9
                ▾ allocationAndRetentionPriority
                  priorityLevelARP: 8
                  pre-emptionCapability: shall-not-trigger-pre-emption (0)
                  pre-emptionVulnerability: not-pre-emptable (0)
```

Network Slicing

UPF configuration by SMF

```
marujo@yoga7i:~/open5GS$ docker exec -it ue3 iperf3 --bind 10.45.0.6 -c 10.40.0.200 -t 10
Connecting to host 10.40.0.200, port 5201
```

```
[ 5] local 10.45.0.6 port 33365 connected to 10.40.0.200 port 5201
```

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd	
[5]	0.00-1.00	sec	896 KBytes	7.33 Mbits/sec	155	2.63 KBytes	
[5]	1.00-2.00	sec	256 KBytes	2.10 Mbits/sec	45	10.5 KBytes	
[5]	2.00-3.00	sec	256 KBytes	2.10 Mbits/sec	39	3.95 KBytes	
[5]	3.00-4.00	sec	256 KBytes	2.10 Mbits/sec	30	1.32 KBytes	
[5]	4.00-5.00	sec	128 KBytes	1.05 Mbits/sec	28	1.32 KBytes	
[5]	5.00-6.00	sec	256 KBytes	2.10 Mbits/sec	26	3.95 KBytes	
[5]	6.00-7.00	sec	128 KBytes	1.05 Mbits/sec	27	1.32 KBytes	
[5]	7.00-8.00	sec	256 KBytes	2.10 Mbits/sec	36	1.32 KBytes	
[5]	8.00-9.00	sec	256 KBytes	2.10 Mbits/sec	34	3.95 KBytes	
[5]	9.00-10.00	sec	384 KBytes	3.14 Mbits/sec	30	3.95 KBytes	

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-10.00	sec	3.00 MBytes	2.52 Mbits/sec	450	sender
[5]	0.00-10.00	sec	2.50 MBytes	2.10 Mbits/sec		receiver

iperf Done.

```
marujo@yoga7i:~/open5GS$ docker exec -it ue3 iperf3 --bind 10.45.0.7 -c 10.40.0.200 -t 10
Connecting to host 10.40.0.200, port 5201
```

```
[ 5] local 10.45.0.7 port 41443 connected to 10.40.0.200 port 5201
```

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd	
[5]	0.00-1.00	sec	135 MBytes	1.13 Gbits/sec	202	158 KBytes	
[5]	1.00-2.00	sec	135 MBytes	1.13 Gbits/sec	231	138 KBytes	
[5]	2.00-3.00	sec	141 MBytes	1.18 Gbits/sec	152	105 KBytes	
[5]	3.00-4.00	sec	129 MBytes	1.08 Gbits/sec	150	107 KBytes	
[5]	4.00-5.00	sec	107 MBytes	898 Mbits/sec	137	93.5 KBytes	
[5]	5.00-6.00	sec	101 MBytes	846 Mbits/sec	161	85.6 KBytes	
[5]	6.00-7.00	sec	130 MBytes	1.09 Gbits/sec	197	94.8 KBytes	
[5]	7.00-8.00	sec	145 MBytes	1.22 Gbits/sec	194	117 KBytes	
[5]	8.00-9.00	sec	97.4 MBytes	817 Mbits/sec	319	80.3 KBytes	
[5]	9.00-10.00	sec	126 MBytes	1.06 Gbits/sec	198	109 KBytes	

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-10.00	sec	1.22 GBytes	1.05 Gbits/sec	1941	sender
[5]	0.00-10.00	sec	1.22 GBytes	1.04 Gbits/sec		receiver

228	376.603558515	10.10.0.4	10.10.0.7	PFCP	638 PFCP Session Establishment Request
229	376.603903186	10.10.0.4	10.10.0.7	PFCP	638 PFCP Session Establishment Request
230	376.603899905	10.10.0.7	10.10.0.4	PFCP	158 PFCP Session Establishment Response
231	376.604114711	10.10.0.7	10.10.0.4	PFCP	158 PFCP Session Establishment Response
232	376.605593016	10.10.0.4	10.10.0.7	PFCP	405 PFCP Session Modification Request
233	376.605919799	10.10.0.7	10.10.0.4	PFCP	86 PFCP Session Modification Response
234	376.608533325	10.10.0.4	10.10.0.7	PFCP	117 PFCP Session Modification Request
235	376.608746824	10.10.0.7	10.10.0.4	PFCP	63 PFCP Session Modification Response
236	376.810770713	10.10.0.4	10.10.0.7	PFCP	117 PFCP Session Modification Request
237	376.810960186	10.10.0.7	10.10.0.4	PFCP	63 PFCP Session Modification Response
238	379.241027548	aa:65:29:ca:b9:7c	5a:0d:6d:7d:22:3d	ARP	42 Who has 10.10.0.7? Tell 10.10.0.4
239	379.241063070	5a:0d:6d:7d:22:3d	aa:65:29:ca:b9:7c	ARP	42 10.10.0.7 is at 5a:0d:6d:7d:22:3d
240	379.841896396	10.10.0.1	239.255.255.250	SSDP	215 M-SEARCH * HTTP/1.1
241	380.842812786	10.10.0.1	239.255.255.250	SSDP	215 M-SEARCH * HTTP/1.1
242	381.843418607	10.10.0.1	239.255.255.250	SSDP	215 M-SEARCH * HTTP/1.1
243	382.843759424	10.10.0.1	224.0.0.251	MDNS	82 Standard query 0x0000 PTR _googlecast
244	382.844139000	10.10.0.1	224.0.0.251	MDNS	82 Standard query 0x0000 PTR _googlecast
245	382.844297976	10.10.0.1	239.255.255.250	SSDP	215 M-SEARCH * HTTP/1.1
246	383.845432657	10.10.0.1	224.0.0.251	MDNS	82 Standard query 0x0000 PTR _googlecast
247	383.846323935	10.10.0.1	224.0.0.251	MDNS	82 Standard query 0x0000 PTR _googlecast
248	384.462569676	10.10.0.1	239.255.255.250	UDP/XML	844 40777 → 3702 Len=802
249	384.516286012	10.10.0.1	239.255.255.250	UDP/XML	844 40777 → 3702 Len=802
250	384.622963359	10.10.0.1	239.255.255.250	UDP/XML	844 40777 → 3702 Len=802

Spare: 0

Node ID : IPv4 address: 10.10.0.4

F-SEID : SEID: 0x000000000000bca, IPv4 10.10.0.4

Create PDR : [Grouped IE]: PDR ID: 1

Create PDR : [Grouped IE]: PDR ID: 2

Create PDR : [Grouped IE]: PDR ID: 3

Create PDR : [Grouped IE]: PDR ID: 4

Create FAR : [Grouped IE]: FAR ID: Dynamic by CP 1

Create FAR : [Grouped IE]: FAR ID: Dynamic by CP 2

Create FAR : [Grouped IE]: FAR ID: Dynamic by CP 3

Create URR : [Grouped IE]: URR ID: Dynamic by CP 1

Create QER : [Grouped IE]: QER ID: Dynamic by CP 1

Create BAR : [Grouped IE]: BAR ID: 1

PDN Type : IPv4

User ID :

APN/DNN : internet

S-NSSAI :

IE Type: S-NSSAI (257)

IE Length: 4

SST: 02

SD: 020000

[Response In: 231]

Frame 231: Packet, 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface upf-cp, id 0

Ethernet II, Src: 5a:0d:6d:7d:22:3d (5a:0d:6d:7d:22:3d), Dst: aa:65:29:ca:b9:7c (aa:65:29:ca:b9:7c)

Internet Protocol Version 4, Src: 10.10.0.7, Dst: 10.10.0.4

User Datagram Protocol, Src Port: 8805, Dst Port: 8805

Packet Forwarding Control Protocol

Flags: 0x21, SEID (S)

Message Type: PFCP Session Establishment Response (51)

Length: 112

SEID: 0x000000000000bca

Sequence Number: 325

Spare: 0

Node ID : IPv4 address: 10.10.0.7

Cause : Request accepted(success)

IE Type: Cause (19)

IE Length: 1

Cause: Request accepted(success) (1)

F-SEID : SEID: 0x0000000000009f1, IPv4 10.10.0.7

Created PDR : [Grouped IE]: PDR ID: 2

Created PDR : [Grouped IE]: PDR ID: 3

Created PDR : [Grouped IE]: PDR ID: 4

[Response To: 229]

[Response Time: 211.525 microseconds]

Propagação do QoS no AMF – UE3 Slice 1

SST:1 SD:000001 (Default S-NSSAI)							
DNN/APN	Type	5QI/QCI	ARP	Capability	Vulnerability	MBR DL/UL	GBR DL/UL
internet	IPv4	9	8	Disabled	Disabled	2 Mbps / 2 Mbps	
		1	2	Enabled	Enabled	10 Mbps / 10 Mbps	5 Mbps / 5 Mbps

PCC Rules

Flow Direction*

Downlink

×

×

Description*

permit out udp from any to 10.40.0.1

Hint: 5.4.2 Flow-Description in TS29.212

+

5QI/QCI*

1

ARP Priority Level (1-15)*

2

MBR Downlink

10

Unit

Mbps

MBR Uplink

10

Unit

Mbps

GBR Downlink

5

Unit

Mbps

GBR Uplink

5

Unit

Mbps

Propagação do QoS no AMF – UE3 Slice 1

No.	Time	Source	Destination	Protocol	Length	Info
29	23.147406007	10.10.0.22	10.10.0.5	NGAP/N...	138	InitialUEMessage, Registration request [RRCEstablishmentCause=mo-Signalling]
87	23.161359954	10.10.0.5	10.10.0.22	NGAP/N...	146	SACK (Ack=0, Arwnd=106496), DownlinkNASTransport, Authentication request
88	23.162302845	10.10.0.22	10.10.0.5	NGAP/N...	146	SACK (Ack=0, Arwnd=106496), UplinkNASTransport, Authentication response
95	23.165987990	10.10.0.5	10.10.0.22	NGAP/N...	126	SACK (Ack=1, Arwnd=106496), DownlinkNASTransport, Security mode command
96	23.166495560	10.10.0.22	10.10.0.5	NGAP/N...	194	SACK (Ack=1, Arwnd=106496), UplinkNASTransport
97	23.166722340	10.10.0.5	10.10.0.22	NGAP	98	SACK (Ack=2, Arwnd=106496), UEContextReleaseCommand [Cause: Nas=normal-release]
99	23.166812606	10.10.0.22	10.10.0.5	NGAP	122	SACK (Ack=2, Arwnd=106496), UEContextReleaseComplete
161	23.171035753	10.10.0.5	10.10.0.22	NGAP/N...	246	SACK (Ack=3, Arwnd=106496), InitialContextSetupRequest
162	23.171133664	10.10.0.22	10.10.0.5	NGAP	98	SACK (Ack=3, Arwnd=106496), InitialContextSetupResponse
165	23.375105328	10.10.0.22	10.10.0.5	NGAP/N...	362	UplinkNASTransport, UplinkNASTransport, UplinkNASTransport
166	23.375819028	10.10.0.5	10.10.0.22	NGAP/N...	154	SACK (Ack=7, Arwnd=106395), DownlinkNASTransport
196	23.394226046	10.10.0.4	10.10.0.5	HTTP2/...	755	DATA[1], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
199	23.394661818	10.10.0.5	10.10.0.22	NGAP/N...	250	PDUSessionResourceSetupRequest
203	23.394912613	10.10.0.4	10.10.0.5	HTTP2/...	753	DATA[3], JSON (application/json), PDU session establishment accept, PDUSessionResourceSetupRequestTransfer
206	23.395250015	10.10.0.5	10.10.0.22	NGAP/N...	234	PDUSessionResourceSetupRequest
209	23.397167582	10.10.0.22	10.10.0.5	NGAP	142	SACK (Ack=6, Arwnd=106496), PDUSessionResourceSetupResponse
211	23.397547850	10.10.0.5	10.10.0.4	HTTP2/...	353	DATA[9], JSON (application/json), PDUSessionResourceSetupResponseTransfer
214	23.398129685	10.10.0.4	10.10.0.5	HTTP2/...	717	DATA[5], JSON (application/json), PDU session modification command, PDUSessionResourceModifyRequestTransfer
217	23.398554211	10.10.0.5	10.10.0.22	NGAP/N...	206	SACK (Ack=8, Arwnd=106496), PDUSessionResourceModifyRequest
221	23.399583200	10.10.0.22	10.10.0.5	NGAP	142	SACK (Ack=7, Arwnd=106496), PDUSessionResourceSetupResponse
223	23.399936703	10.10.0.5	10.10.0.4	HTTP2/...	353	DATA[11], JSON (application/json), PDUSessionResourceSetupResponseTransfer

```

▼ Item 0: id-QosFlowAddOrModifyRequestList
  ▼ ProtocolIE-Field
    id: id-QosFlowAddOrModifyRequestList (135)
    criticality: reject (0)
  ▼ value
    ▼ QosFlowAddOrModifyRequestList: 1 item
      ▼ Item 0
        ▼ QosFlowAddOrModifyRequestItem
          qosFlowIdentifier: 2
          ▼ qosFlowLevelQosParameters
            ▼ qosCharacteristics: nonDynamic5QI (0)
              ▼ nonDynamic5QI
                fiveQI: 1
            ▼ allocationAndRetentionPriority
              priorityLevelARP: 2
              pre-emptionCapability: may-trigger-pre-emption (1)
              pre-emptionVulnerability: pre-emptable (1)
          ▼ gBR-QosInformation
            maximumFlowBitRateDL: 10000000bits/s
            maximumFlowBitRateUL: 10000000bits/s
            guaranteedFlowBitRateDL: 5000000bits/s
            guaranteedFlowBitRateUL: 5000000bits/s

```



DEMONSTRAÇÃO DE QOS